

The objective of this article is to introduce some of the tools that can be used when carrying out these tasks.

Learning management systems (LMS)

An LMS is an important tool for making the learning process effective. It includes various tasks such as administration, the delivery of e-content material, and reporting. These days, LMS is popular both in higher education and in corporate training. Learning management systems are Web based tools to assist the e-learning process. There are both proprietary and open source LMSs available in the market. Some of the open source options are listed below:

- Moodle
- Sakai
- ATutor
- OpenOLAT
- Canvas

The following are some of the proprietary tools:

- Blackboard
- Edmodo
- Taleo
- eCollege
- Kannu

This article introduces readers to two FOSS LMSs – Sakai and ATutor.

Sakai: Sakai is an open source LMS. The first version of it was released in 2005. The latest stable release is version 19.0, which came out in March 2019. Sakai is available with the Educational Community Licence. The core of the Sakai LMS is written in Java. It has support for 19 different languages. Sakai is a community based project with major contributions from the Apereo Foundation – a non-profit organisation. The Sakai project is the result of a grant from the Mellon Foundation. Major institutions in the US such as Indiana University, Georgia Institute of Technology, Massachusetts Institute of Technology, Stanford University and the University of Michigan have adopted the early

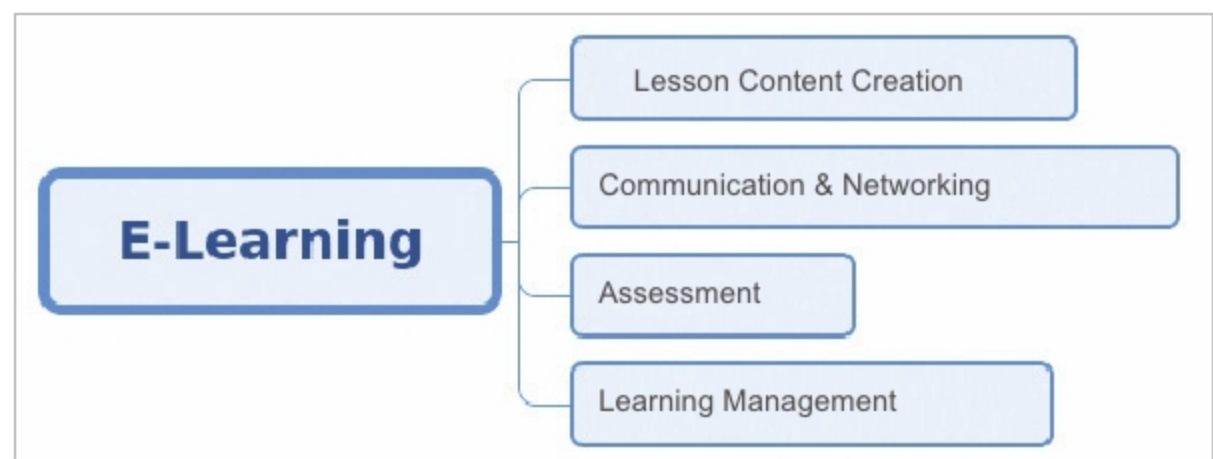


Figure 1: Associated tasks of the e-learning content creation and delivery process

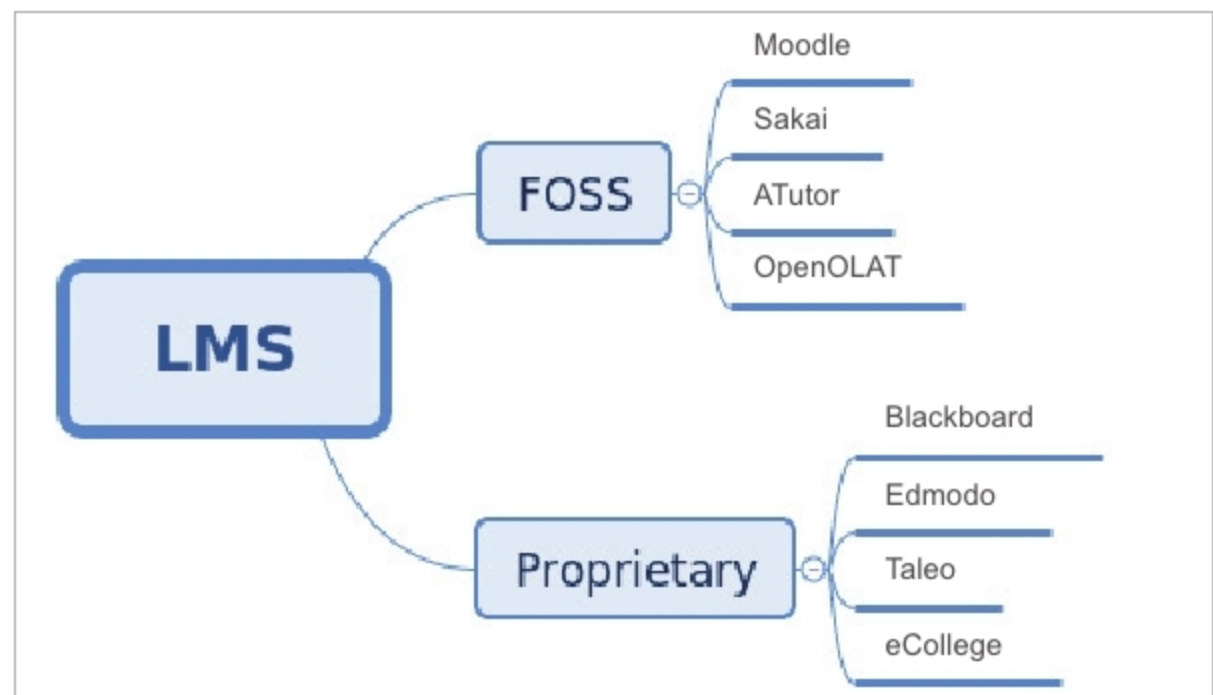


Figure 2: Open source and proprietary LMSs

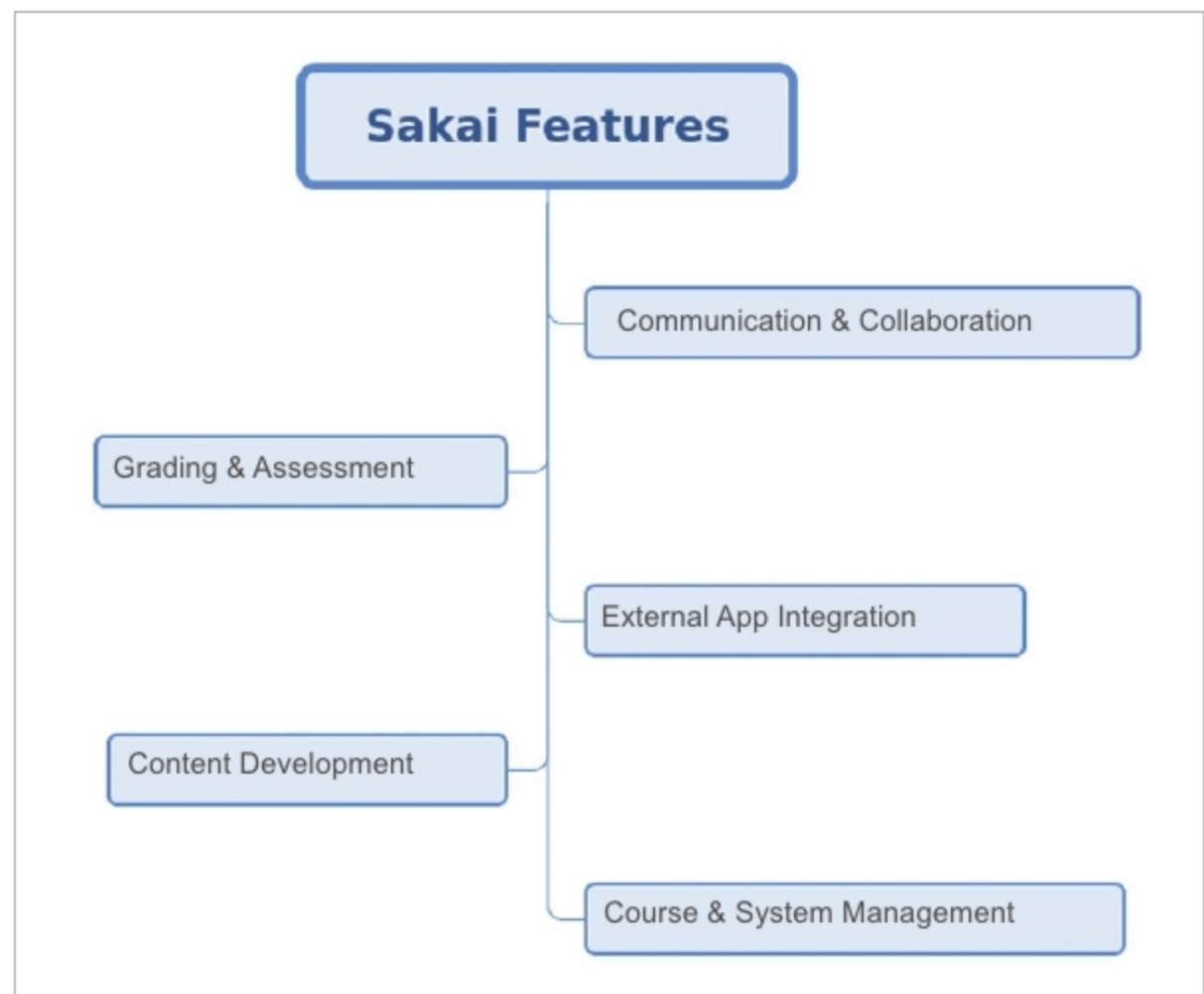


Figure 3: Sakai's features